

Apurva Umredkar

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SUMMARY

Graduate student in Computer Science with 2 years of enterprise software engineering experience and deep specialization in applied AI/ML, including generative and agentic AI. Proficient in building scalable machine learning systems from prototype to deployment, with strong technical communication and cross-functional collaboration skills.

EXPERIENCE

AI Product Development Intern, *Vaspian LLC*

09/2025 – Present

- Gained practical experience in integrating MCP servers, deploying containers with Docker, and managing version control using Git.
- Developed a RAG-powered FAQ chatbot using n8n workflows and Bluemesh AI Studio, leveraging prompt engineering, structured output generation, and a knowledge base built from call transcripts.
- Created a conversational AI agent in Eleven Labs and integrated it with Airtable using webhooks to automate the management of webinar invitations and customer response tracking.

SAP ABAP Technical Consultant, *Capgemini Technology Services India Limited*

08/2022 – 07/2024

- Integrated external systems with SAP BRIM/SOM, serving over 10,000 users and reducing order processing time by 40% through scalable REST APIs using SAP ABAP, ABAP CDS, and BRF+.
- Collaborated with the data migration team to map over 500 fields across 5 production systems and led system integration tests for more than 25 interfaces alongside the security and QA teams.
- Resolved an average of 8-10 ServiceNow tickets weekly, encompassing production bug analysis and feature implementation through technical investigation and code development in SAP ACM.

EDUCATION

- Master's, Computer Science (AI/ML), University at Buffalo (GPA: 3.78/4)

08/ 2024 – Present

- Bachelor's, Electronics & Communication, National Institute of Technology Nagpur

07/2018 – 05/2022

PROJECTS

On-device Agentic AI Assistant *VLMs, Ollama, MCP Servers, LangGraph, SQLite, ChromaDB*

- Designed and implemented a local, privacy-focused agentic chatbot leveraging agentic orchestration frameworks and iterative Plan-Action-Reflect loops to enable dynamic task planning, visual reasoning, and tool invocation without external dependencies.
- Developed a multi-tier memory architecture using for short, episodic, and long-term memory retention, integrating Spotify, Notion, Google Workspace, internal file system and web search MCP servers to extend agent capabilities while ensuring on-device execution.

UBAssist: RAG-powered University Chatbot *LLMs, RAG, Cursor, Hugging Face, Docker, FAISS, LoRA, QLoRA*

- Developed an end-to-end AI chatbot using a fine-tuned Microsoft Phi 1.5 model with parameter-efficient training techniques, achieving 48% faster training and 12 percentage point improvement in answer accuracy.
- Built knowledge base of 18,000+ university documents with hybrid search capabilities, enabling real-time responses while reducing incorrect answers by 30% compared to baseline models.
- Deployed scalable web application using Flask, FastAPI, and Docker on cloud infrastructure, optimizing model to run efficiently with only 13M parameters while maintaining performance.

Grayscale to RGB Image Colorization *CNN, UNet, Vision Transformers, GANs, Gradio*

- Built complete machine learning workflow using U-Net architecture augmented with Vision Transformer (ViT) blocks, attention gates, and patch discriminator-based adversarial training, achieving 94.15% structural similarity with ground truth images on test set of 5,000 images.
- Improved model color fidelity using a weighted loss function for rare color classes and patch-level GAN loss to enforce realism and texture consistency.
- Deployed as an interactive Gradio application on Hugging Face Spaces, enabling real-time image colorization with sub-second inference time.

PUBLICATIONS

“Improved search and rescue optimization based load balancing scheme of CCTV footage on edge computing environment” *Concurrency and Computation: Practice and Experience*, 2023. [DOI](#)

- Conducted Monte Carlo simulations to analyze the impact of video and network parameters on YOLO CNN-based edge video processing latency, emulating multi-device setups using ngrok tunneling.

SKILLS

Programming & Data: Python, C++, SQL, NumPy, Pandas, Matplotlib, Seaborn, Plotly, PostgreSQL, MySQL, SQLite

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, FAISS, ChromaDB, Qdrant, Hugging Face, Ollama, Context Engineering, n8n, MCP Servers, Google ADK

Backend & DevOps: FastAPI, Docker, Git, GitHub Actions